TASALIGN: AN OBJECTIVE EVALUATION METHOD FOR TEXT-AUDIO SEMANTIC ALIGNMENT BASED ON MIXTURE OF EXPERTS AND SEQUENTIAL CROSS-ATTENTION

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ABSTRACT

Text-to-audio (TTA) technology has wide applications in media and virtual reality, yet evaluating its semantic alignment—alongside audio quality—remains challenging. While modern TTA systems generate high-fidelity audio, their semantic fidelity is often weak. Traditional evaluation relies on subjective judgments, which are costly, inconsistent, and unscalable. To address this, we propose an objective semantic alignment evaluator based on a Mixture of Experts (MoE) architecture with Sequential Cross-Attention (SeqCoAttn).

Our method uses RoBERTa and BEATs as text and audio encoders, respectively, with projection heads for cross-modal dimension alignment. SeqCoAttn models bidirectional temporal correspondences to produce fused representations. The MoE integrates four specialized experts—covering general alignment, multi-scale audio analysis, universal audio-language representation, and fine-grained temporal matching—whose predictions are dynamically weighted by a gating network. Training employs a hybrid loss combining Clipped MSE and a margin-based contrastive loss to improve robustness and ranking performance.

Evaluated on a standard dataset (7.5K train / 3K validation pairs), our model outperforms baselines across all metrics, achieving an SRCC of 0.668—a 28.4% relative improvement—demonstrating its effectiveness as a scalable, reliable objective evaluator for TTA semantic alignment.

Index Terms— text-to-audio, semantic alignment evaluation, mixture of experts, cross-modal feature fusion, objective evaluation metric

1. INTRODUCTION

Text-to-audio (TTA) technology, which automatically synthesizes audio samples from textual descriptions has garnered significant attention in recent years due to its broad application prospects in media content production, virtual reality audio environment creation, and other fields [1]. Evaluating TTA systems typically involves two core dimensions: audio quality and text-audio relevance. While modern TTA models have achieved remarkable progress in generating high-quality audio, they often fail to fully reflect the content of input text, leading to a critical gap in relevance performance [2]. Text-audio relevance—defined as the degree to which synthesized audio matches the semantic content of the input text—is typically evaluated subjectively. However, human assessment is time-consuming, expensive, and suffers from low inter-test comparability, hindering scalability in large-scale TTA development. This underscores the urgent need for an objective metric that closely correlates with human judgments.

While recent advances in generative AI have greatly improved audio quality, semantic misalignment remains a critical bottleneck: models often produce high-fidelity audio that contradicts the text (e.g., generating "bird chirping" for "heavy rain"), undermining real-world usability.

Existing objective methods [3] typically employ a single cross-modal model, struggling to simultaneously capture general alignment, fine-grained semantics, and temporal dynamics. They also lack robustness under imbalanced score distributions (e.g., datasets dominated by high-quality samples). To address these limitations, we propose a novel objective evaluator based on a Mixture-of-Experts (MoE) architecture enhanced with Sequential Cross-Attention (SeqCoAttn). Our approach:

- Uses RoBERTa and BEATs as encoders with aligned projection heads, and SeqCoAttn to model bidirectional audio–text temporal correspondence;
- Integrates four specialized experts—covering general alignment, multi-scale audio analysis, universal representation learning, and fine-grained temporal matching—with dynamic gating for adaptive fusion;
- Employs a hybrid loss combining Clipped MSE and a margin-based contrastive loss to improve robustness against data imbalance and annotation noise;
- Demonstrates state-of-the-art performance on a standard benchmark (7.5K/3K train/validation pairs), achieving an SRCC of 0.668—a 28.4

This work provides an efficient, reliable, and scalable solution for objective TTA evaluation and offers new directions for cross-modal generation assessment.

2. DATASETS

2.1. Overview

We mainly use the officially provided dataset to train the model. The dataset consists of the following contents.

- **Audio-text pairs:** It contains original real audio and synthesized audio.
- **Subjective semantic-alignment scores:** 11-point semantic-alignment scores between audio and text per listener
- **Average semantic-alignment scores.** These are the average scores of each audio-text pair.
- **Listener IDs:** A string identifier of the number of listeners who rated the audio-text pair

2.2. Analysis of dataset

We first conducted an overall inspection of the officially provided data and found that there are a total of 7500 pairs of training data and 3000 pairs of validation data, and all audio samples have been converted into mono 16-bit 16kHz audio. The training data and validation data have completely different listeners. In addition, since the semantic alignment score for each audio-text pair ranges between 0 and 10, we grouped all training and validation data based on this score. The distribution of these grouped data can be observed in Figure 1. The histogram reveals a significant imbalance in score distribution, characterized by a heavy concentration of high-score samples and severe scarcity of low-score instances.

3. PROPOSED APPROACH

3.1. Basic Architecture

As shown in Fig 2, the basic model architecture is detailed as follows. The architecture integrates specialized encoders for each modality, a cross-attention fusion mechanism, and a final projection layer for regression.

3.1.1. Audio-Text Encoder

The text encoder is responsible for extracting contextualized text features from input captions. It employs a pre-trained transformer-based language model (RoBERTa [4]). To align the text features with the audio modality for subsequent fusion, a projection head is employed to map them into the same dimensional space as the audio features. For efficiency and to leverage transfer learning, the base language model is partially frozen, with only the parameters of the top two transformer layers being fine-tuned during training.

The audio encoder processes raw waveform inputs to generate discriminative audio features. It utilizes a pre-trained BEATs [5] model as its backbone. A novel weighted sum aggregation strategy is employed, where learnable weights are assigned to the output of each encoder layer, allowing the model to dynamically emphasize informative layers. The aggregated features are then projected into the same dimensional space using a projection head similar to that of the text encoder. Similarly, the BEATs encoder is partially frozen during training, with only the parameters of its top two layers being updated.

3.1.2. SeqCoAttn

To effectively fuse the audio and text sequences, a cross-attention mechanism is employed. This module consists of two separate Multi-Head Attention (MHA) layers:

- An audio-to-text attention layer that computes attention weights over the text sequence using the audio sequence as queries, enabling the audio features to attend to relevant textual information.
- A text-to-audio attention layer that computes attention weights over the audio sequence using the text sequence as queries, allowing the text features to focus on pertinent audio segments.

After cross-attention, both the audio and text sequences are independently pooled using adaptive max-pooling to produce fixed-dimensional vectors. These vectors are then concatenated to form the final fused representation.

3.2. Mixture of Experts

We employs a Mixture of Experts (MoE) architecture to leverage the complementary strengths of multiple specialized models for the task of predicting the average semantic alignment score of synthesized audio-text pairs. The MoE framework dynamically combines the predictions of several expert models through a gating mechanism, allowing for more nuanced and accurate estimation.

3.2.1. Expert Models

The system consists of four different pre-trained expert models, each fine-tuned for this challenge, and they differ in their architecture and input methods:

- Expert 1 adopts the pretrained CLAP [6] model as its backbone, which is rooted in large-scale contrastive

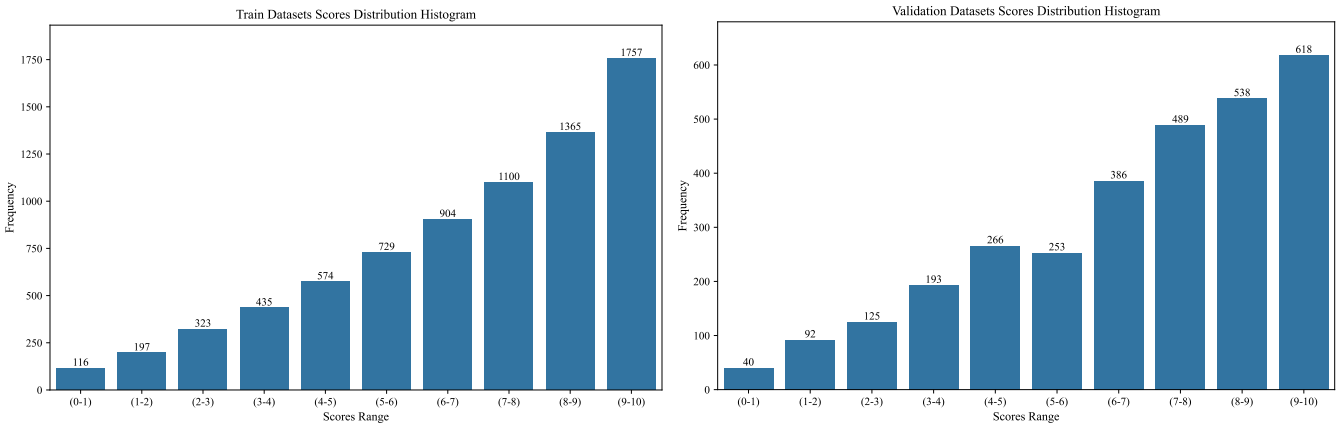


Fig. 1: Scores Distribution of Datasets (left: training set; right: validation set).

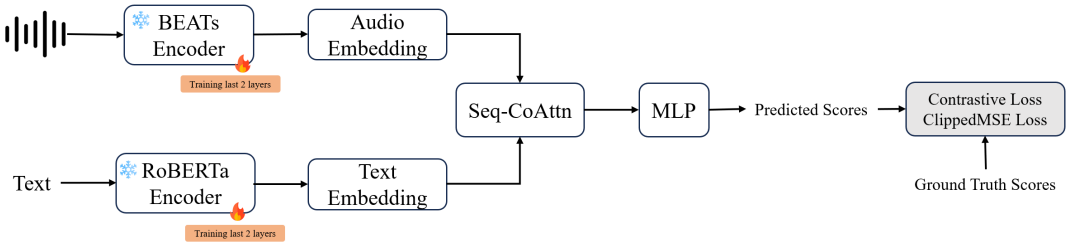


Fig. 2: Basic Architecture

learning on audio-text pairs (e.g., the LAION-Audio-630K dataset) . Its core architecture follows a dual-encoder design: it employs general-purpose audio encoders and text encoders to project audio and text into a shared latent space, achieving cross-modal semantic alignment via contrastive loss . It excels at general cross-modal tasks such as zero-shot audio classification and basic text-to-audio retrieval, demonstrating strong generalization on standard benchmarks . In the MOE framework, Expert 1 specializing in providing generalizable cross-modal semantic cues to complement other experts

- Expert 2 is based on MGA-CLAP [7], an extension of CLAP that integrates Multi-Scale Guided Attention (MGA) to capture audio features across temporal scales. The MGA module enhances audio representation by jointly modeling short-term details (e.g., transients, noise) and long-term semantics (e.g., emotion, content), improving discriminability without compromising text–audio alignment. Within the MoE system, Expert 2 specializes in fine-grained audio analysis (particularly for complex or mixed-sound scenes) complementing Expert 1's general cross-modal capabilities.

- Expert 3 is built on M2D-CLAP [6], a two-stage framework that combines CLAP's cross-modal contrastive learning with the self-supervised Masked Modeling Duo (M2D) to overcome CLAP's limited audio generalizability. In the first stage, M2D learns universal audio representations via masked prediction; in the second, CLAP aligns these with LLM-derived text embeddings and refines cross-modal features under M2D's guidance. This yields robust, transferable audio-language representations—particularly beneficial for regression tasks like MOS prediction, where CLAP's discrete language grounding falls short. Within the MoE system, Expert 3 bridges pure audio understanding and cross-modal alignment, complementing Expert 1's general cross-modal capability and Expert 2's multi-scale audio focus.

- Expert 4 employs a custom cross-modal architecture designed for fine-grained audio–text alignment. It integrates a BEATs-based audio encoder, a transformer-based text encoder (RoBERTa), and a sequence-level cross-attention (SeqCoAttn) fusion module to model temporal correspondence between modalities, addressing the global-pooling limitations of other experts.

  The audio branch extracts layer-wise features from a pre-trained BEATs model, combines them via learnable weights, and projects them to a 512-dimensional

space. The text branch generates contextualized token embeddings and aligns them to the same dimension. SeqCoAttn then applies bidirectional cross-attention: audio queries attend to relevant text tokens, and vice versa. The resulting sequences are pooled via adaptive max-pooling, concatenated, and fed into a two-layer MLP projection head—with dropout and optional Tanh clipping—to produce the alignment scores.

Within the MoE framework, Expert 4 specializes in capturing local temporal alignments between audio and text, complementing Experts 1–3's strengths in general cross-modal matching, multi-scale audio modeling, and universal representation learning, respectively.

All four expert models are fixed during the training of the MoE model. Their weights are frozen, meaning they do not undergo further parameter updates. This is done to preserve the specialized knowledge each expert has acquired during its individual pretraining phase.

3.2.2. Gating Network

The Gating Network is responsible for adaptively combining the predictions of the four experts. Its role is to determine the confidence assigned to each expert's output for a given input. Unlike the expert models, the Gating Network is trainable. During training, it learns to assign appropriate weights to each expert based on the input characteristics, ef- fectively determining which expert(s) are most relevant for predicting the alignment score of a particular audio-text pair.

3.3. Loss function

We adopt the same loss formulation as the baseline, combining clipped Mean Squared Error (MSE) and a contrastive loss via weighted fusion. The contrastive loss is defined as a margin-based bounded error on score differences:

$$\mathcal{L}_{\text{con}} = \max(0, \left|\hat{d} - d\right| - \varepsilon)$$

where $\hat{d}$ and d are the ground-truth and predicted score differences between sample pairs. This loss penalizes only when the deviation exceeds a margin ε, prioritizing correct ranking over absolute accuracy. Crucially, it imposes a strong penalty when the signs of d and $\hat{d}$ disagree (i.e., when the model reverses the relative relevance order) thereby enhancing discrimination of alignment degrees and improving rank-based metrics like SRCC. The clipped MSE loss is given by:

$$\mathcal{L}_{\text{ClippedMSE}} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(|\hat{y} - y_i| > \tau) \cdot (\hat{y} - y_i)^2$$

where s_i and $\hat{s}_i$ denote the ground-truth and predicted scores for the i-th utterance, respectively, and $\tau > 0$ is a clipping threshold that limits the contribution of large errors. This formulation mitigates the undue influence of outliers or poorly calibrated annotations, thereby improving training stability and robustness. The final loss function is defined as: $\mathcal{L} = \beta \cdot \mathcal{L}_{\text{mse}} + \gamma \cdot \mathcal{L}_{\text{con}}$ where β and γ are hyperparameters.

4. EXPERIMENTAL SETUPS

We empirically chose τ = 0.25 for clipped MSE loss, α = 0.1 for contrastive loss, $\beta_{cbl} = 0.99$ for CBL, and $\beta = 1.0$ and $\gamma = 0.5$ for the weights of the two loss functions. The batch size was 8. Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) was used as the optimizer with an initial learning rate of 1.0×10^{-5}. Also, learning rate scheduling with linear warm-up and linear decay was used. The optimal model was selected referring to Spearman's rank correlation coefficient (SRCC) calculated from the validaion set.

5. RESULT

The model's performance across various metrics on the validation set is presented in Table 1. It is evident that our model consistently outperforms existing baseline approaches across all evaluated metrics, achieving a notable 28.4% relative improvement in SRCC.

System	SRCC ↑	LCC ↑	KTAU ↑	MSE ↓
Ours	**0.668**	**0.680**	**0.486**	**3.334**
Baseline	0.384	0.264	0.396	4.836

Table 1: Performance comparison between our model and the baseline on the validation set.

6. CONCLUSION

To address the inefficiency and inconsistency of subjective evaluation in text-to-audio (TTA) semantic alignment, we propose an objective evaluator based on Mixture of Experts (MoE) and Sequential Cross-Attention (SeqCoAttn).

Our method first identifies the imbalanced score distribution in TTA datasets (few low-quality samples), then builds a cross-modal encoder using RoBERTa and BEATs, enhanced by SeqCoAttn for fine-grained temporal alignment. Four specialized experts—covering general alignment, multi-scale audio analysis, universal representation, and temporal matching—are dynamically fused via a gating network. A hybrid loss (Clipped MSE + contrastive) improves robustness and ranking accuracy.

On the validation set, our model achieves SRCC = 0.668, a 28.4% relative gain over baselines, confirming strong alignment with human judgment. This provides a scalable, reliable solution for TTA evaluation and supports semantic-aware model development.

7. REFERENCES

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